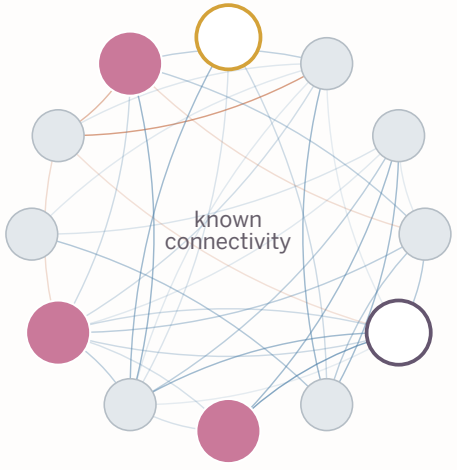
Prior-averaged risk often leaves recording decisions unchanged

Record 3 units to predict an integrated stimulus response: known connectivity, uncertain physiology; simulated 12-unit networks



recorded node (M = 3)

stimulated node

response readout

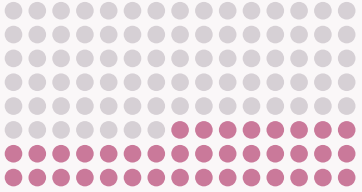
$$q = \int [r_{\text{stim}}(t) - r_{\text{control}}(t)]\, dt$$

Blue: positive; orange: negative connections

1 Recording agreement

Nominal vs prior-averaged risk
40 independent networks

68% 82 of 120 single-target designs are identical



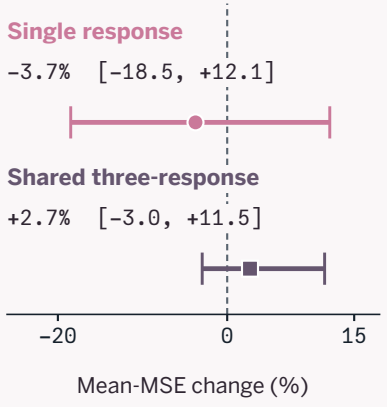
65% 26 of 40 shared-target designs are identical



same recording set

2 Response error

Common MAP fit and budget
40 independent networks



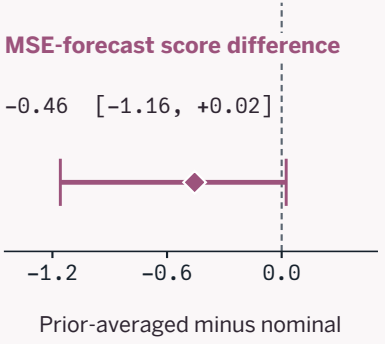
97.5% paired network-cluster CI

3 Risk forecasts

Fixed data and point prediction
40 new independent networks

19 of 40 networks score lower; 21 score higher

One network supplies 64% of the summed score difference



95% paired network-cluster CI

With identical recordings, the common estimator gives identical predictions.

All primary response-error and risk-forecast intervals include zero.

Exploratory 100-unit family: 4 of 8 intervals exclude zero, all favouring prior averaging (8 independent networks).

Extension: unadjusted 95% intervals; one point estimate reverses sign. All quantities are simulated.